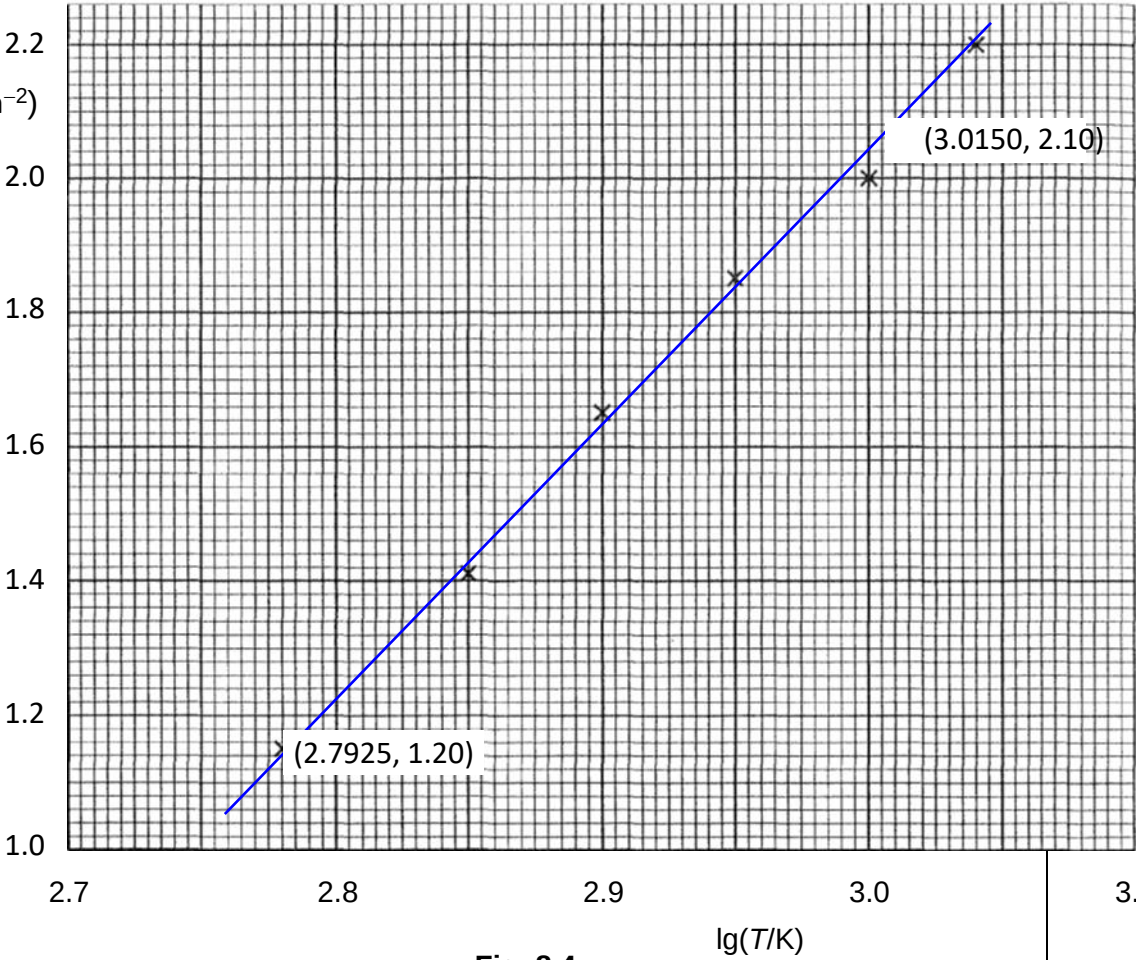


Qns	Answer				Marks																					
8(a)(i)	Label V between 400-700 nm on the horizontal axis.				B1																					
8(a)(ii)	Wavelengths corresponding to red light would have the greatest intensity within the spectrum of visible light.				B1																					
8(b)(i)	<table><tr><td>T / K</td><td>$\lambda_{\text{max}} / \text{nm}$</td><td>$T \times \lambda_{\text{max}}$</td></tr><tr><td>600</td><td>4830</td><td>2898000</td></tr><tr><td>700</td><td>4140</td><td>2898000</td></tr><tr><td>800</td><td>3610</td><td>2888000</td></tr><tr><td>900</td><td>3210</td><td>2889000</td></tr><tr><td>1000</td><td>2900</td><td>2900000</td></tr><tr><td>1100</td><td>2630</td><td>2893000</td></tr></table>				T / K	$\lambda_{\text{max}} / \text{nm}$	$T \times \lambda_{\text{max}}$	600	4830	2898000	700	4140	2898000	800	3610	2888000	900	3210	2889000	1000	2900	2900000	1100	2630	2893000	B1
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	Since the product $T \times \lambda_{\text{max}}$ for different temperatures are the same when rounded to 2 s.f., it is a constant.				B1																					
	$\text{constant} = (T \times \lambda_{\text{max}})_{\text{average}} = 2.9 \times 10^6 \text{ K nm}$				B1																					
	Note: Unit for constant is required in A level examination.																									
8(b)(ii)	$T \times \lambda_{\text{max}} = 2.89 \times 10^6$ $\lambda_{\text{max}} = \frac{2.89 \times 10^6}{1200}$ $= 2.41 \times 10^3 \text{ nm} = 2.41 \times 10^{-6} \text{ m}$				M1 A1																					

8(c)(i) $\lg(I_{\text{tot}}/\text{W m}^{-2})$	 Fig. 8.4	3.1
	$I_{\text{tot}} = cT^n$ $\lg I_{\text{tot}} = n \lg T + \lg c$ Hence, n is the gradient of the $\lg I_{\text{tot}}$ against $\lg T$ graph. $\text{gradient} = \frac{2.10 - 1.20}{3.0150 - 2.7925} = 4.045$ $n = 4.045$	M1 M1 A1
8(c)(ii)	$I_{\text{tot}} = cT^n$ $71 = c(900)^{4.045}$ $c = 7.97 \times 10^{-11}$ I_{tot} at 1200 K $= 7.97 \times 10^{-11} \times 1200^{4.045} = 227 \text{ W m}^{-2}$	M1 M1 A1
8(d)	The curve should be similar to the curves for the other temperatures. It should start from the origin and peak at 2410 nm. It should be above the 1100 K curve (except intersecting at the origin). Graph has to be smooth. Note: Peak should be more pointed than that for 1100 K.	B1 B1 B1

8(e)(i)	Either wavelength λ_{max} (wavelength at maximum intensity) or I_{tot} (total intensity) Determine the temperature using $T = \frac{\text{constant}}{\lambda_{\text{max}}}$ or $I_{\text{tot}} = cT^n$	B1 B1
8(e)(ii)	advantage: No physical contact is needed to make the measurement. disadvantage: (any of the followings) The intensity varies smoothly with wavelength. It is difficult to measure λ_{max} to high precision. Low intensity with low temperature, difficult to measure. Presence of background radiation.	B1 B1